

**NATALIE R. MYERS, PhD**

**Address:** Johns Hopkins University (JHU)  
Department of Physics & Astronomy  
Baltimore, MD 21218

**Email:** [nrmyers.astro@gmail.com](mailto:nrmyers.astro@gmail.com)  
**Orcid ID:** 0000-0001-9738-4829  
**Website:** [nrmyers.github.io](https://nrmyers.github.io)

### Education:

|      |                      |                                                                                                                                                                                                                         |
|------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>Ph.D. Physics</b> | Astrophysics Track, Texas Christian University (TCU), Fort Worth, TX<br><i>“Tracing the Chemical Enrichment of the Milky Way Using Star Clusters: Measurement and Analysis of Neutron Capture Elemental Abundances”</i> |
| 2022 | <b>M.S. Physics</b>  | Texas Christian University (TCU), Fort Worth, TX<br><i>“Measuring Galactic Chemical Abundance Trends in Open and Globular Clusters Using SDSS/APOGEE”</i>                                                               |
| 2020 | <b>B.S. Physics</b>  | University of St. Thomas (UST), St. Paul, MN (Minor: Math)                                                                                                                                                              |

### Employment and Affiliation:

2025–Present    **Assistant Research Scientist,** Johns Hopkins University, *Supervisor:* K. Schlaufman  
2021–2025    **Research Assistant,** Texas Christian University, *Advisor:* P. M. Frinchaboy  
2020–2025    **Teaching Assistant,** Texas Christian University  
2019–2020    **Physics Tutor,** University of St. Thomas  
2018–2020    **Observatory Crew Member,** University of St. Thomas

## Teaching Experience

### TA and Tutoring Experience:

|             |             |                                         |              |
|-------------|-------------|-----------------------------------------|--------------|
| 2025 Spring | PHYS 10164: | Classical Physics 2 Lab                 | (2 sections) |
| 2024 Spring | PHYS 10164: | Classical Physics 2 Lab                 | (3 sections) |
| 2021 Summer | PHYS 10164: | Classical Physics 2 Lab                 | (1 section)  |
| 2021 Spring | PHYS 10263: | Cosmic Origins Lab                      | (2 sections) |
| 2020 Fall   | PHYS 10273: | Intro to Astronomy: Earth & Planets Lab | (2 sections) |
| 2019-2020   | Tutoring:   | Up to Calculus III and Modern Physics   |              |

### Workshops and Guest Lectures:

|             |             |                                                                                                                                                |
|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 Summer |             | John's Hopkins Teaching Academy: Teaching Institute<br><i>Pedagogy-focused; including active learning, course design, effective assessment</i> |
| 2026 Spring | AS.171.611: | Stellar Structure and Evolution lecture<br><i>Covered graduate level lecture on nucleosynthesis</i>                                            |
| 2023 Fall   | PHYS 10273: | Cosmic Origins guest lecture: Potential for Life in the Solar System                                                                           |
| 2023 Fall   | PHYS 30123: | Observational Physics Lab Assistant (2 nights)<br><i>Trained 2.7-m Telescope during class trip to McDonald Obs.</i>                            |

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### Leadership Experience

#### Co-Mentoring Graduate Students

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|-----------|--------------------------|--------------------------------------------------------|
| 2025–2026 | R. Bachchan (TCU)        | Taught spectroscopic methods and how to use BACCHUS    |
| 2024–2026 | A. Sinha (Univ. of Utah) | Assisted in BACCHUS analysis training                  |
| 2024      | J. Otto (TCU)            | Trained on the 2.7-m telescope at McDonald Observatory |
| 2023      | A. Wiggins (TCU)         | Trained on the 2.7-m telescope at McDonald Observatory |

#### Mentoring Undergraduate Students

|           |                                                                                        |  |
|-----------|----------------------------------------------------------------------------------------|--|
| 2026      | Mentored Johns Hopkins Rowland Summer Research Fellow: J. Mendo                        |  |
| 2023      | Supervised REU student: H. Wallace                                                     |  |
| 2023      | Co-mentored REU student: K. Toguchi-Tani                                               |  |
| 2022      | Co-Mentored REU student: K. Thomas                                                     |  |
| 2018–2019 | Observatory Worker                                                                     |  |
|           | – Trained introductory astronomy students to operate UST 0.4-m telescope (4 semesters) |  |
|           | – Gave tours for star parties and other community outreach events                      |  |

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### Grants, Awards, and Fellowships:

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|-----------|-----------------------------------------------------------------------------------|
| 2026      | Cool Stars 23 Poster Award (I was one of 3 awardees out of ~ 330 posters)         |
| 2026      | AAS International Travel Grant (\$2049)                                           |
| 2024      | AAS International Travel Grant (\$998)                                            |
| 2024      | Sigma Xi Grant in Aid of Research (\$2,765)                                       |
| 2024      | Graduate Travel Grant, TCU (\$400, \$800)                                         |
| 2023      | AAS 241 Chambliss Astronomy Achievement Award Winner (Poster)                     |
| 2020–2021 | Ida Green Fellowship, TCU (\$56,010)                                              |
| 2019–2020 | MN Space Grant Consortium Award for Significant Contributions to Research (\$500) |
| 2019      | Young Scholars Research Grant, Advisor: M. Wood                                   |

#### Observing Experience:

|        |                                 |                                    |
|--------|---------------------------------|------------------------------------|
| 2026   | Hubble Space Telescope Cycle 34 | (Co-I; 28 orbits)                  |
| 2026Q2 | APO 3.5-m/ARCES                 | (PI; 2 half nights)                |
| 2026Q1 | APO 3.5-m/ARCES                 | (PI; 2 half nights)                |
| 2024B  | McDonald 2.7-m/Coudé            | (Co-I; 3 nights)                   |
| 2024A  | Keck 10-m/HIRES                 | (Co-I; 9 half nights)              |
| 2023C  | McDonald 2.7-m/Coudé            | (PI; 6 nights)                     |
| 2023C  | McDonald 2.7-m/Coudé            | (PI; 6 nights)                     |
| 2023B  | Magellan 6.5-m/MIKE             | (Co-I; 2 nights)                   |
| 2023B  | Lick 2.4-m/APF                  | (Co-I; 7 nights , queue scheduled) |
| 2023B  | Lick 3.0-m/Hamilton echelle     | (Co-I; 3 nights)                   |
| 2023B  | Keck 10-m/HIRES                 | (Co-I; 1 night )                   |
| 2023B  | McDonald 2.7-m/Coudé            | (Co-I; 4 nights)                   |
| 2023A  | Magellan 6.5-m/MIKE             | (Co-I; 2 nights)                   |
| 2022B  | McDonald 2.7-m/Coudé            | (Co-I; 4 nights)                   |
| 2022A  | Magellan 6.5-m/MIKE             | (Co-I; 2 nights)                   |

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### Other Professional Experience and Volunteering

- Volunteer at the JHU Annual Physics Fair (Spring 2026)
  - TAC committee member (2026)
  - Astronomy & Astrophysics Journal Referee (2025,2026)
  - Graduate Student Representative at TCU (2023-2024)
  - CUWiP Conference Local Organizing Committee (LOC) member and volunteer, (Jan 2023 at TCU)
  - Volunteering for Atatiana Projects 2023 STEAM summer camp (July 2023)
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### Research Interests

I am an astronomer with a particular interest in Galactic Chemical Evolution, nucleosynthesis, and stellar evolution. In particular, I have worked to constrain star cluster membership (Myers et al. 2022, Schiavon et al. 2024), derive neutron-capture abundances and characterize how those elements vary by location in the Milky Way (Myers et al. *submitted*), and analyze age-chemistry relationships for stellar populations (Myers et al. *in prep*).

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### Technical Skills

Python • Data Science • MATLAB • Microsoft Office • OOP • Science Communication  
GitHub • Spectral Analysis • LaTeX • Observational Astronomy • SQL

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## NATALIE R. MYERS – PUBLICATIONS

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### Refereed Publications

- SDSS Collaboration, Adamane Pallathadka, G., Aghakhanloo, M., Aird, J., Almeida, A., et al. (212 authors listed alphabetically; including **Myers, N.**), (2025), “*The Nineteenth Data Release of the Sloan Digital Sky Survey*”, *Astronomical Journal*, *submitted*, doi: 10.48550/arXiv.2507.07093.
- Sinha, A., Zasowski, G., **Myers, N. R.**, Manea, C., Frinchaboy, P., et al., (2026), “*The Chemical Homogeneity of Single-Lined Spectroscopic Binaries in Open Clusters*”, *MNRAS*, 547, 440, doi: 10.1093/mnras/stag440.
- Otto, J. M., Frinchaboy, P. M., **Myers, N. R.**, Johnson, J. W., Donor, J., et al., (2025), “*The Open Cluster Chemical Abundances and Mapping Survey: VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*”, *Astronomical Journal*, 171, 91, doi: 10.3847/1538-3881/ae28d8.
- Kollmeier, J. A., Rix, H.-W., Aerts, C., Aird, J., Alfaro, P. V., et al. (220 authors total, including **Myers, N.**), (2025), “*Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy*”, *Astronomical Journal*, 171, 52, doi: 10.3847/1538-3881/ae0576.
- Spoo, T., Thomas, K., “Kaleo” Toguchi-Tani, E., Otto, J., **Myers, N.**, et al., (2025), “*Extending the [C/N]-Age Calibration: Using Globular Clusters to Explore Older and Metal-Poor Populations*”, *Astronomical Journal*, 170, 333, doi: 10.3847/1538-3881/ae14ec.
- Schiavon, R. P., Phillips, S. G., **Myers, N.**, Horta, D., Minniti, D., et al., (2024), “*The APOGEE value-added catalogue of Galactic globular cluster stars*”, *MNRAS*, 528, 1393, doi: 10.1093/mnras/stad3020.
- Almeida, A., Anderson, S. F., Argudo-Fernández, M., Badenes, C., Barger, K., **et al.** (155 authors listed alphabetically; including **Myers, N.**), (2023), “*The Eighteenth Data Release of the Sloan Digital Sky Surveys: Targeting and First Spectra from SDSS-V*”, *ApJS*, 267, 44, doi: 10.3847/1538-4365/acda98.
- Myers, N.**, Donor, J., Spoo, T., Frinchaboy, P. M., Cunha, K., et al., (2022), “*The Open Cluster Chemical Abundances and Mapping Survey. VI. Galactic Chemical Gradient Analysis from APOGEE DR17*”, *Astronomical Journal*, 164, 85, doi: 10.3847/1538-3881/ac7ce5.
- Spoo, T., Tayar, J., Frinchaboy, P. M., Cunha, K., **Myers, N.**, et al., (2022), “*The Open Cluster Chemical Abundances and Mapping Survey. VII. APOGEE DR17 [C/N]-Age Calibration*”, *Astronomical Journal*, 163, 229, doi: 10.3847/1538-3881/ac5d53.
- Abdurro’uf, Accetta, K., Aerts, C., Silva Aguirre, V., Ahumada, R., **et al.** (341 authors listed alphabetically; including **Myers, N.**), (2022), “*The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data*”, *ApJS*, 259, 35, doi: 10.3847/1538-4365/ac4414

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## Talks

*Open Clusters in the APOGEE Era: Stellar Laboratories, Galactic Probes, and a Survey that Changed the Game*, (Aug. 2026), APOGEE’s Galactic Odyssey, Teruel, Spain.

**\*\*Invited Talk!**

*No Universal Chemical Clock: Testing Age–Abundance Relationships with Accurate and Precise Subgiant Ages*, (Jun. 2026), Cool Stars 23 Splinter, Tokyo, Japan.

*Characterizing Galactic Neutron Capture Gradients with Open Clusters*, (Jun. 2026), Cool Stars 23 Poster Award, Tokyo, Japan.

*Understanding the Chemical Composition of the Milky Way: A Study of Open Clusters and the Heavy Elements Within*, (Nov. 2025), Invited Speaker, CAS Emerging Scholars Seminar, University of St. Thomas, St. Paul, MN.

**\*\*Invited Talk!**

*Neutron Capture Abundances of SDSS-IV/APOGEE-Selected Open Cluster Stars: Galactic Gradients and Applications to Chemical Clocks*, (Sep.2025), Wine and Cheese Seminar, Johns Hopkins University, Baltimore, MD.

*Neutron Capture Abundances of SDSS-IV/APOGEE-Selected Open Cluster Stars: Galactic Gradients and Applications to Chemical Clocks*, (Jan. 2025), AAS245 dissertation talk, 129.06D, National Harbor, MD.

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## Conference Proceedings

**Myers, N.**, Loebman, S., Reggiani, H., & Frinchaboy, P., (2025), “*Beyond OCCAM: Measuring Optical Neutron Capture Abundances of Open Cluster Stars*”, Proceedings of the IAU Symposium 395, doi: 10.48550/arXiv.2510.13014.

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## Select Poster Presentations

**Myers, N.**, Frinchaboy, P., Loebman, S., Reggiani, H., Donor, J., Shetrone, M., (2026), “*Characterizing Galactic Neutron Capture Gradients with Open Clusters*”, Cool Stars 23, Tokyo, Japan, June 2026  
**\*\* Poster Award Winner!**

**Myers, N.**, Frinchaboy, P., Loebman, S., Reggiani, H., Holmbeck, E., Donor, J., Shetrone, M., (2024), “*Neutron Capture Follow-up of SDSS-IV/APOGEE Open Clusters: Galactic Gradients and Applications to Chemical Clocks*”, IAU Symposium 395: “Stellar populations in the Milky Way and beyond”, Paraty, Brazil, November 2024

**Myers, N.**, Reggiani, H., Loebman, S., Holmbeck, E., Shetrone, M., Frinchaboy, P., & The Occam Team, (2024), “*Surveying for Ancient Supernovae Across the Galaxy: Mapping SDSS-IV/OCCAM Abundances and the Search for Heavier Elements*”, AAS, 243, 458.07, January 2024.

**Myers, N.**, Frinchaboy, P., Donor, J., Spoo, T., Otto, J., Wiggins, et al., (2023), “*Mapping Chemical Abundances Across the Milky Way with Open Clusters in SDSS/APOGEE: Gradients of Iron and Light Elements plus Neutron Capture Follow-Up*”, Surveying the Milky Way: The Universe in Our Own Backyard, Pasadena CA, October 2023.

**Myers, N.,** Frinchaboy, P., Spoo, T., Wiggins, A., Otto, J., et al., (2023), “*The Open Cluster Chemical Abundances and Mapping Survey Follow-up: Tracing the Neutron-Capture Enrichment of the Milky Way*”, SDSS-V Meeting NYC, August 2023.

**Myers, N.,** ”Painting a Portrait of a Young Milky Way using Globular Clusters”, (2023), Student Research Symposium at TCU, April 2023.

**Myers, N.,** Frinchaboy, P., Donor, J., & Spoo, T., (2023), “*Exploring the Milky Way’s Childhood: Globular Clusters Abundances from SDSS/APOGEE*”, AAS, 241, 402.14.

**\*\* Chambliss Award Winner!**

**Myers, N.,** Frinchaboy, P., Donor, J., Cunha, K., Spoo, T., et al., (2022), “*OCCAM VI: Assembling the open cluster “Avengers” of Galactic Evolution*”, AAS, 240, 201.04, January 2022.

**Myers, N., & Wood, M.,** (2021), “*Calibrating a New Spectrometer Using Arcturus*”, AAS, 237, 547.18, January 2021 (Virtual due to COVID-19).

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## Collaborative Research Presentations

Spoo, T., Frinchaboy, P., Tayar, J., Souto, D., & The Occam Team, (2024), “*Deriving Ages for Field Stars: Tools and Complications from Stellar Evolution*”, AAS, 243, 430.04D.

Otto, J., Spoo, T., Toguchi-Tani, E., Thomas, K., Frinchaboy, P., & The Occam Team, (2024), “*Digging through the Galactic Graveyard: Chemistry and Ages of “Dead” Milky Way Satellite Galaxies*”, AAS, 243, 107.05.

Wiggins, A., Bhattarai, B., Loebman, S., **Myers, N.,** Frinchaboy, P., Otto, J., Spoo, T., & Donor, J., (2024), “*Only the Special Survive: Star Cluster Disruption in Milky Way-like Galaxy Simulations*”, AAS, 243, 458.20.

Wallace, H., **Myers, N.,** Otto, J., Donor, J., Spoo, T., Wiggins, A., & Frinchaboy, P., (2024), “*The Open Cluster Chemical Abundances and Mapping (OCCAM) Survey: First Results from SDSS-V*”, AAS, 243, 458.02.

Wiggins, A. I., Bhattarai, B., **Myers, N.,** Loebman, S., Frinchaboy, P., et al., (2023), “*Only the Special Survive: Star Cluster Disruption in Galaxy Simulations*”, Surveying the Milky Way: The Universe in Our Own Backyard, Pasadena CA.

Otto, J., Wallace, H., Frinchaboy, P., **Myers, N.,** Spoo, T., et al., (2023), “*The Open Cluster Chemical Abundance and Mapping (OCCAM) survey: Cross-calibration of APOGEE to MWM/ASTRA*”, New York City NY.

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## Other Conferences Attended

2023 Conference for Undergraduate Women in Physics at Texas Christian Univ., Fort Worth, TX.

2020 Conference for Undergraduate Women in Physics at Univ. of Minnesota, St. Paul, MN.

2019 Conference for Undergraduate Women in Physics at Northwestern Univ., Chicago, IL.

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